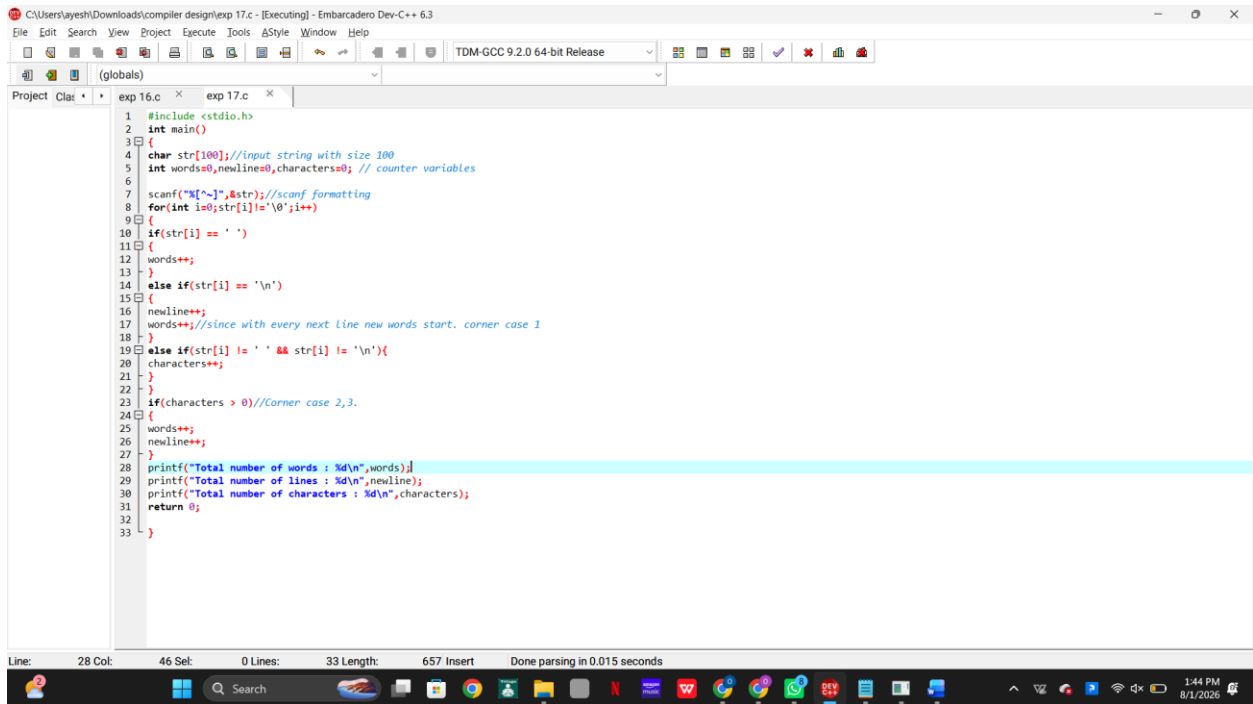


Exp 17:

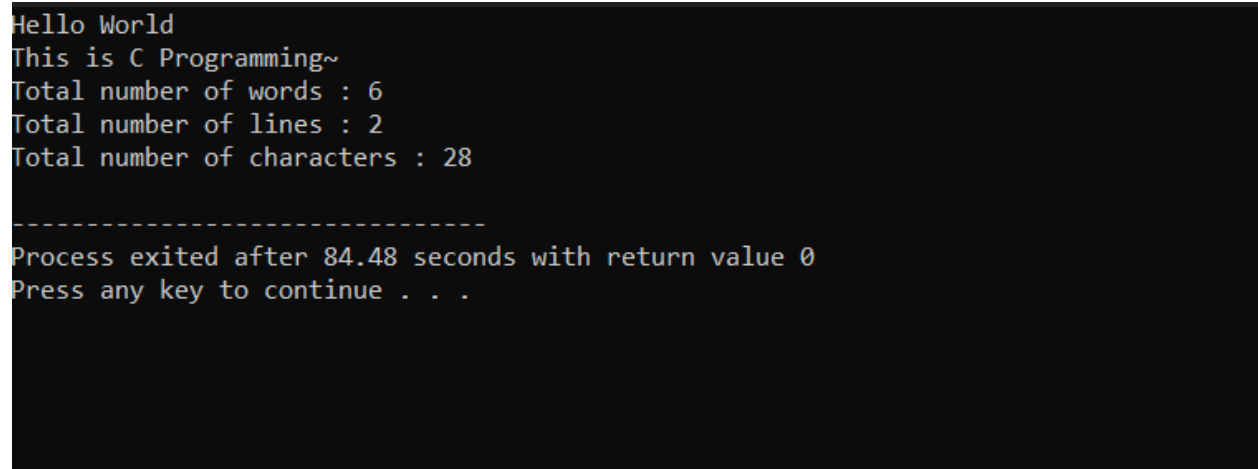
Code:



The screenshot shows a code editor window titled "C:\Users\ayesh\Downloads\compiler design\exp 17.c - [Executing] - Embarcadero Dev-C++ 6.3". The code is a C program that counts words, lines, and characters in a string. The string is "Hello World\nThis is C Programming~\nTotal number of words : 6\nTotal number of lines : 2\nTotal number of characters : 28". The program uses a loop to iterate through the string, counting words (separated by spaces), lines (separated by newlines), and characters. It then prints the counts and returns 0.

```
1 #include <stdio.h>
2 int main()
3 {
4     char str[100]; //input string with size 100
5     int words=0, newline=0, characters=0; // counter variables
6     scanf("%[^\n]", &str); //scanf formatting
7     for(int i=0; str[i]!='\0'; i++)
8     {
9         if(str[i] == ' ')
10        {
11            words++;
12        }
13        else if(str[i] == '\n')
14        {
15            newline++;
16            words++; //since with every next line new words start. corner case 1
17        }
18        else if(str[i] != ' ' && str[i] != '\n'){
19            characters++;
20        }
21    }
22    if(characters > 0) //Corner case 2,3.
23    {
24        words++;
25        newline++;
26    }
27    printf("Total number of words : %d\n", words);
28    printf("Total number of lines : %d\n", newline);
29    printf("Total number of characters : %d\n", characters);
30    return 0;
31 }
```

Output:



The screenshot shows the output of the program. It prints the counts for words, lines, and characters, followed by a separator line and the process exit message.

```
Hello World
This is C Programming~
Total number of words : 6
Total number of lines : 2
Total number of characters : 28

-----
Process exited after 84.48 seconds with return value 0
Press any key to continue . . .
```